

Tut 5

Ex. 9.6

Q36

$$\sum_{n=1}^{\infty} \frac{\cos(n\pi)}{n}$$

Important:

$$\cos(n\pi) = (-1)^n \text{ for } n \geq 1$$

$$\frac{\cos(n\pi)}{n} = \frac{(-1)^n}{n} \rightarrow \text{study } \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$



We can introduce $a_n, b_n, \dots$

for convenience.

Exp. $b_n = \frac{1}{n}$

$$\Rightarrow \frac{(-1)^n}{n} = (-1)^n b_n$$

• Look at absolute series

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n} \quad (\text{div.})$$

• Verify that

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n} \text{ conv. by } \underline{\underline{AST}}$$

→ Mention all the hypotheses

① $\frac{1}{n} \geq 0$

② $\left\{ \frac{1}{n} \right\}$ decreasing

③ $\frac{1}{n} \rightarrow 0$

• (conclude) The series conv. cond.

Sol. Let $a_n = (-1)^{n+1} \frac{1+n}{n^2}$ for $n \geq 1$.

Consider $|a_n| = \frac{1+n}{n^2}$. $\left[\frac{1+n}{n^2} \sim \frac{1}{n} \right]$

(LCT) Let $b_n = \frac{1}{n}$ for $n \geq 1$.

$$\lim_{n \rightarrow \infty} \frac{|a_n|}{b_n} = \lim_{n \rightarrow \infty} \frac{\frac{1+n}{n^2}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{1+n}{n} = \lim_{n \rightarrow \infty} \left(\frac{1}{n} + 1 \right) = 1 > 0$$

$$\left[\because \frac{1}{n} \rightarrow 0 \right]$$

Since $\sum_{n=1}^{\infty} \frac{1}{n}$ div., the series $\sum_{n=1}^{\infty} |a_n|$ div. by LCT.

$$\begin{aligned}\textcircled{3} \lim_{n \rightarrow \infty} \frac{1+n}{n^2} &= \lim_{n \rightarrow \infty} \frac{1}{n^2} + \frac{1}{n} \\ &= 0 + 0 \quad \left[\because \frac{1}{n^2} \rightarrow 0, \frac{1}{n} \rightarrow 0 \right] \\ &= 0\end{aligned}$$

By A.S.T, $\sum_{n=1}^{\infty} a_n$ conv.

$\Rightarrow \sum_{n=1}^{\infty} a_n$ conv. conditionally.

Ex. 9.8E

Q29 $\frac{x}{3} \ln(1+x^2)$

$n=1$ $n=2$ $n=3$ $n=4$

$$\ln(1+x) = \frac{x^1}{1} - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \dots$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n$$

$$[x \rightarrow x^2 \text{ in } \ln(1+x)]$$

$$\therefore \ln(1+x^2) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^{2n}$$

$$\frac{x}{3} \ln(1+x^2) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{3n} x^{2n+1}$$